

WASP-2b

R-0286

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_230623 · created 2026-07-17T16:52:05+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460118.8928 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.114 +/- 0.013
Transit depth (Rp/Rs)^2	1.31 +/- 0.29 [%]
Orbital Inclination (inc)	87.63 +/- 1.33
Airmass coefficient 1 (a1)	0.9787 +/- 0.0071
Airmass coefficient 2 (a2)	0.019 +/- 0.023
Scatter in the residuals of the lightcurve fit is	0.72 %
Best Comparison Star	#3 - [106, 81]
Optimal Method	PSF photometry
Transit Duration (day)	0.0899 +/- 0.0063

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.114 ± 0.013 vs 0.1326 (deviation 1.43σ)
Duration fit vs published	0.0899 d vs 0.0737 d (deviation 2.56σ)
Mid-time O–C	2.1 min
Depth SNR	8.8
Residual scatter	0.72 %

Light curve

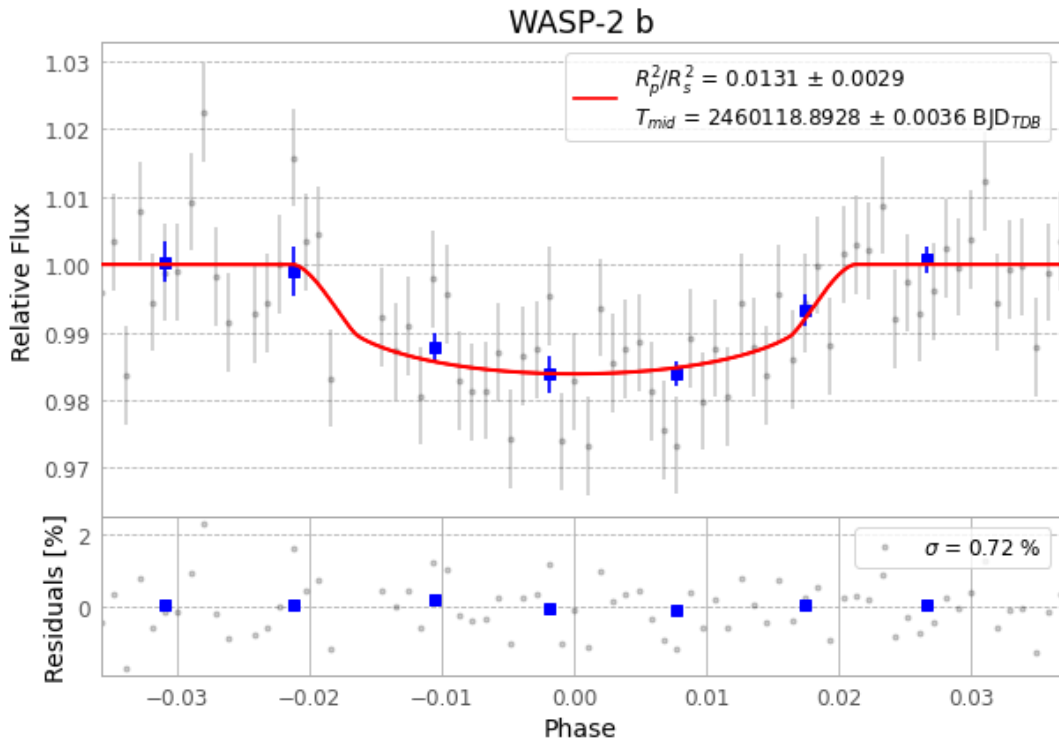


Figure 1 — FinalLightCurve_WASP-2 b_2023-06-23.png (SHA-256 38a91d76d492b0cf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [326, 196], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8c3aa2010b23ec37...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit faa81f0

Data provenance

76 FITS frames (50 MB), WASP-2230623072716.FITS → WASP-2230623111215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-230623.log (da1eabab8c02...), FinalLightCurve_WASP-2 b_2023-06-23.png (38a91d76d492...), FinalLightCurve_WASP-2 b_2023-06-23.csv (9ce53d75cc20...)

Compute resources

Wall time 145.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 16:52Z.